Chloe Barker

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EDUCATION

Southern Methodist University | Dallas, TX Jan 2025 - Aug 2026
M.S. in Data Science | GPA: 3.95

The University of Texas at Austin | Austin, TX Aug 2020 - May 2024
B.S.A. in Biology | Minor: Health Communication | Pre-Health Professions Certificate | GPA: 3.77

TECHNICAL SKILLS

Programming & Analysis: Python (pandas, NumPy, scikit-learn), SQL, R (tidyverse, caret), SAS, PySpark
Machine Learning & Statistics: Regression, classification, logistic regression, LASSO, random forest, gradient boosting, support vector machines, KNN, clustering, time series forecasting, NLP, feature engineering, cross-validation, hyperparameter tuning, model evaluation, hypothesis testing, ANOVA, A/B testing
Data Platforms & Management: Databricks, Snowflake, MySQL, data pipelines, ETL, data cleaning and transformation, relational database design, query optimization
Visualization: Tableau, Matplotlib, seaborn, Plotly, ggplot2, R Shiny
Generative AI & Development: LLM applications, RAG, prompt engineering, Git/GitHub, HTML, CSS
Business & Productivity: Excel (PivotTables, data analysis), PowerPoint, executive presentations, technical documentation

EXPERIENCE

Goosehead Insurance | Westlake, TX May 2026 - Aug 2026
Data Science Intern
- Developed and evaluated machine-learning models using large-scale insurance data in Snowflake and Databricks to support lead prioritization, customer segmentation, and marketing decision-making.
- Built data preparation and modeling workflows using SQL, Python, and PySpark; compared classification approaches and translated analytical findings into recommendations for business stakeholders.

Applied Analytics Group, LLC | Remote Jan 2026 - May 2026
Data Science Analyst (Volunteer)
- Supported early-stage analytics projects using Python, R, and SQL, spanning data cleaning, exploratory analysis, and reporting.
- Standardized and documented technical workflows to improve reproducibility, clarity, and handoff across the team.

Outlier AI | Remote Aug 2024 - Jan 2026
AI Math Evaluator
- Assessed AI-generated mathematical solutions across algebra, calculus, and statistics, documenting reasoning errors and grading outputs for correctness, clarity, and completeness.
- Created structured test cases and error analyses to strengthen the reliability of reasoning-based AI systems.

RUSH University, Dept. of Microbial Pathogens & Immunity | Chicago, IL Summers 2021 & 2023
Research Assistant
- Investigated alcohol, circadian, and HIV-medication effects on the gut microbiome and bone density using longitudinal clinical data in R and Excel.
- Prepared, visualized, and documented experimental data through reproducible workflows for multidisciplinary biomedical research teams.

SELECTED PROJECTS

M.S. Capstone: Groundwater Lithium & Dementia Prevalence | Python
- Modeled dementia prevalence across 254 Texas counties using groundwater and dementia data, engineering multi-year exposure measures and validating results through robustness and sensitivity checks.
- Visualized county-level geographic patterns and presented findings to faculty; received the MSDS Best Capstone Presentation award.

LLM/RAG Document Q&A Application | Databricks, Python
- Created a retrieval-augmented application that answers questions about PDF documents by retrieving relevant passages and generating source-grounded responses.
- Implemented an interactive interface with document processing, retrieval, prompting, and secure API-key management in Databricks.

Diabetes Readmission Triage | Python, scikit-learn
- Developed a risk-ranking pipeline across 71,518 hospital encounters to prioritize patients for limited transitional-care capacity while preventing data leakage.
- Benchmarked models with capacity-aware metrics and translated results into staffing scenarios and stakeholder recommendations.